

openSUSE Leap 16

A Case Study

Timothy Saxena (24261A05R4)

V Harshini (24261A05R5)

Vanam Sai Mani Akshat (24261A05R6)

The Name, Pronunciation & Mascot

- Pronounced: "open-SOO-zuh" — rhymes with open source
- **SUSE** is a German acronym: *Software und System-Entwicklung* (Software & Systems Development)
- Founded in **Nuremberg, Germany** — the name honours **Konrad Zuse**, inventor of the first programmable computer
- "open"**SUSE** — "open" added in 2006 when development was opened to the public community
- Mascot: **Geeko the Chameleon** — symbolises adaptability across any environment



What is openSUSE Leap?

- A free, **open-source** Linux distribution
- Built for **stability, reliability, and professional use**
- Two flavours under openSUSE:
 - **Leap** — stable, fixed releases, You leap from v15 → v16
 - **Tumbleweed** — rolling release, always latest
- Based on **SUSE Linux Enterprise (SLE)** — the OS large corporations pay for
- Current release: **Leap 16** — released October 1, 2025



 **Enterprise-grade Linux, available to everyone for free.**

 **Think of it like:**

Fortune 500 companies pay thousands for SUSE Linux Enterprise. Leap gives you the **same codebase** — for free.

Why Was It Made? The Problem It Solved

Problem 1 – SUSE Was Closed (pre-2005)

SUSE Linux was a paid product only. Community developers could not contribute.

Problem 2 – openSUSE Existed but Was Unstable (2005–2015)

When **development opened**, the community version chased the latest software. Exciting, but too **unreliable** for serious or professional use.

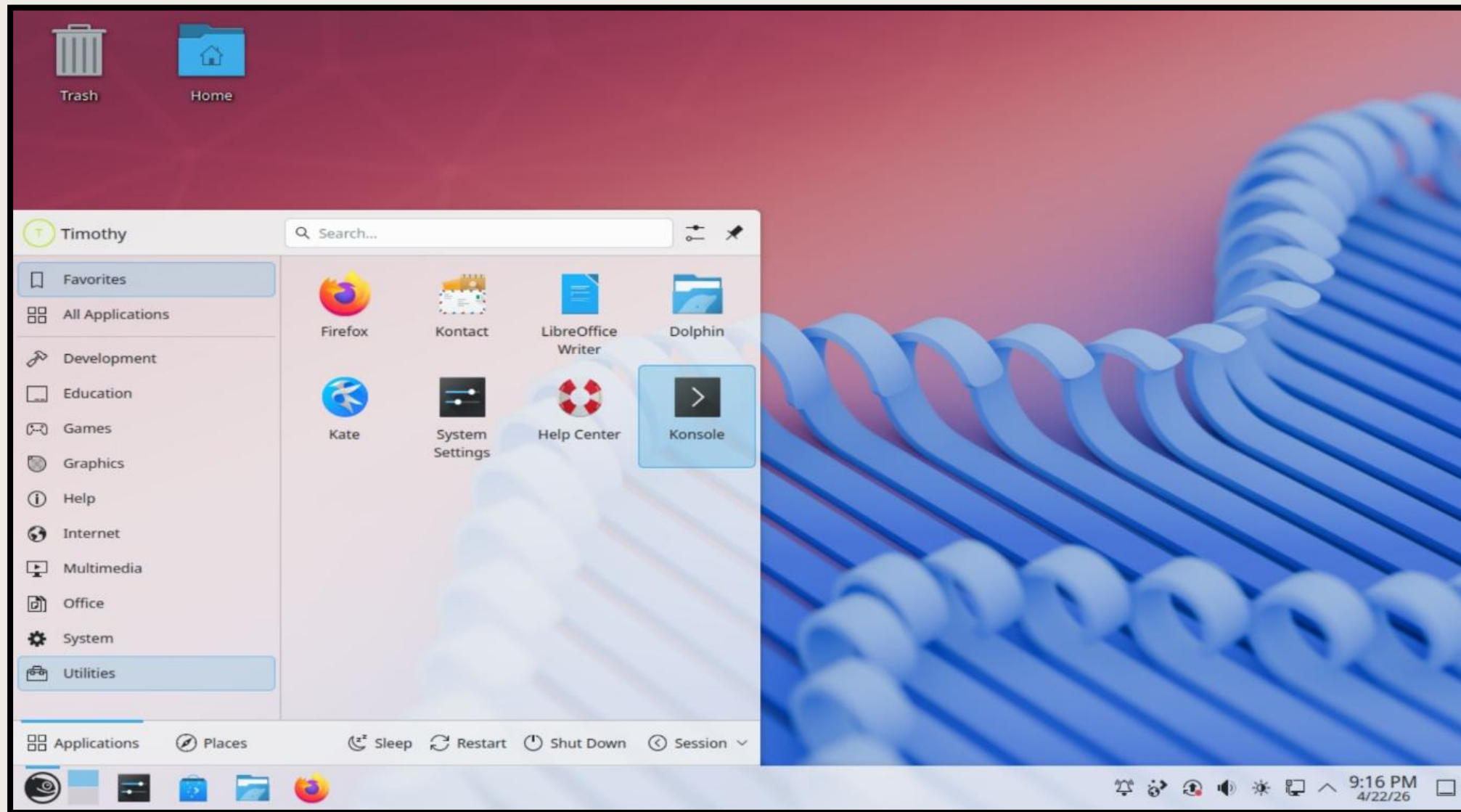
The Solution – Leap (2015):

-  Sync openSUSE directly with SUSE Linux Enterprise source code. Same stability. Same security. Zero cost.

Key Functionality 1 – User Interface

User Interface based on KDE Plasma

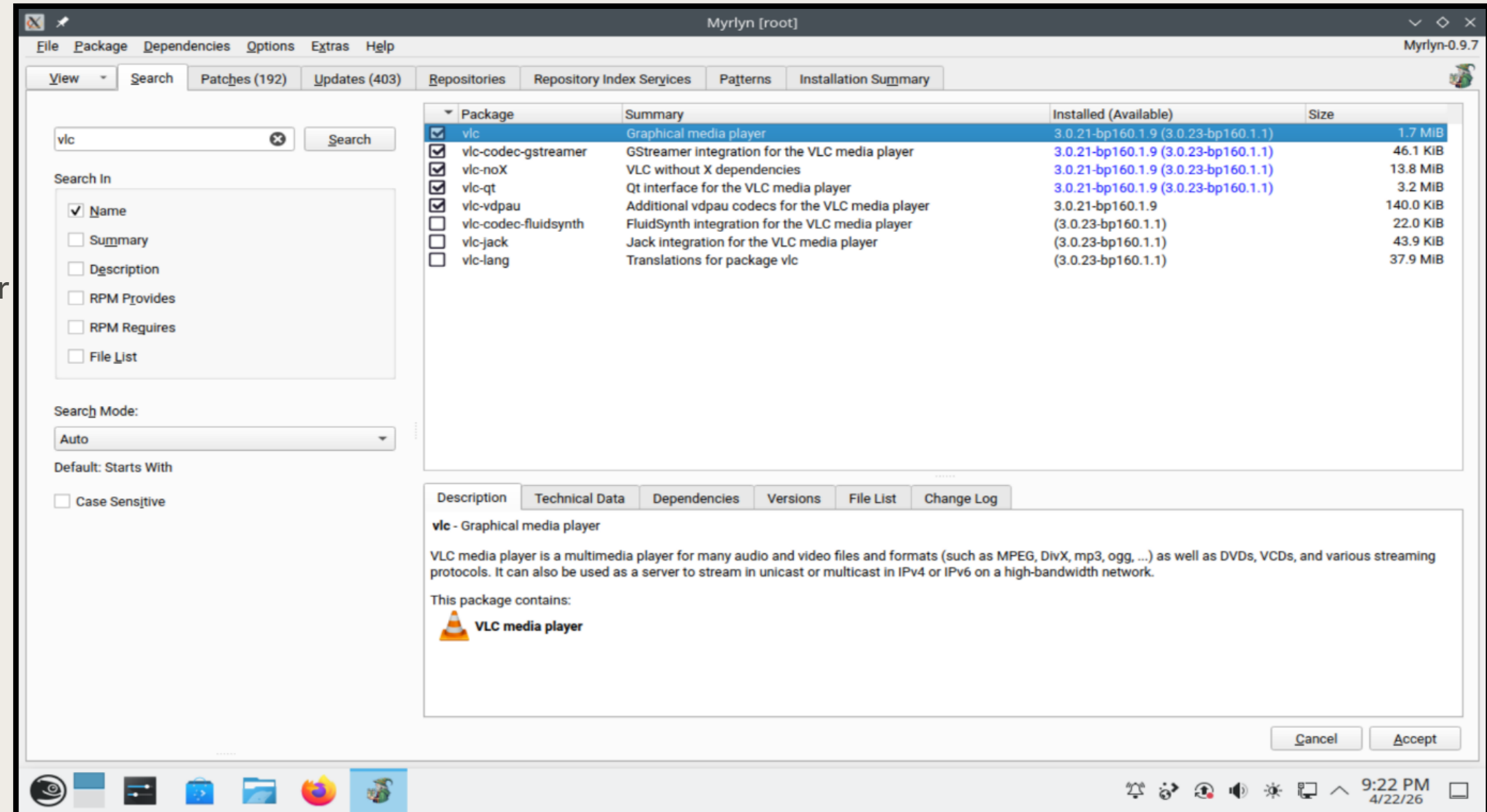
- It gives you the taskbar, file manager, settings app, window decorations, start menu – all of it.
- openSUSE Leap 16 offers KDE Plasma 6.4 as the main desktop choice



Key Functionality 2 – Installing & Managing Software

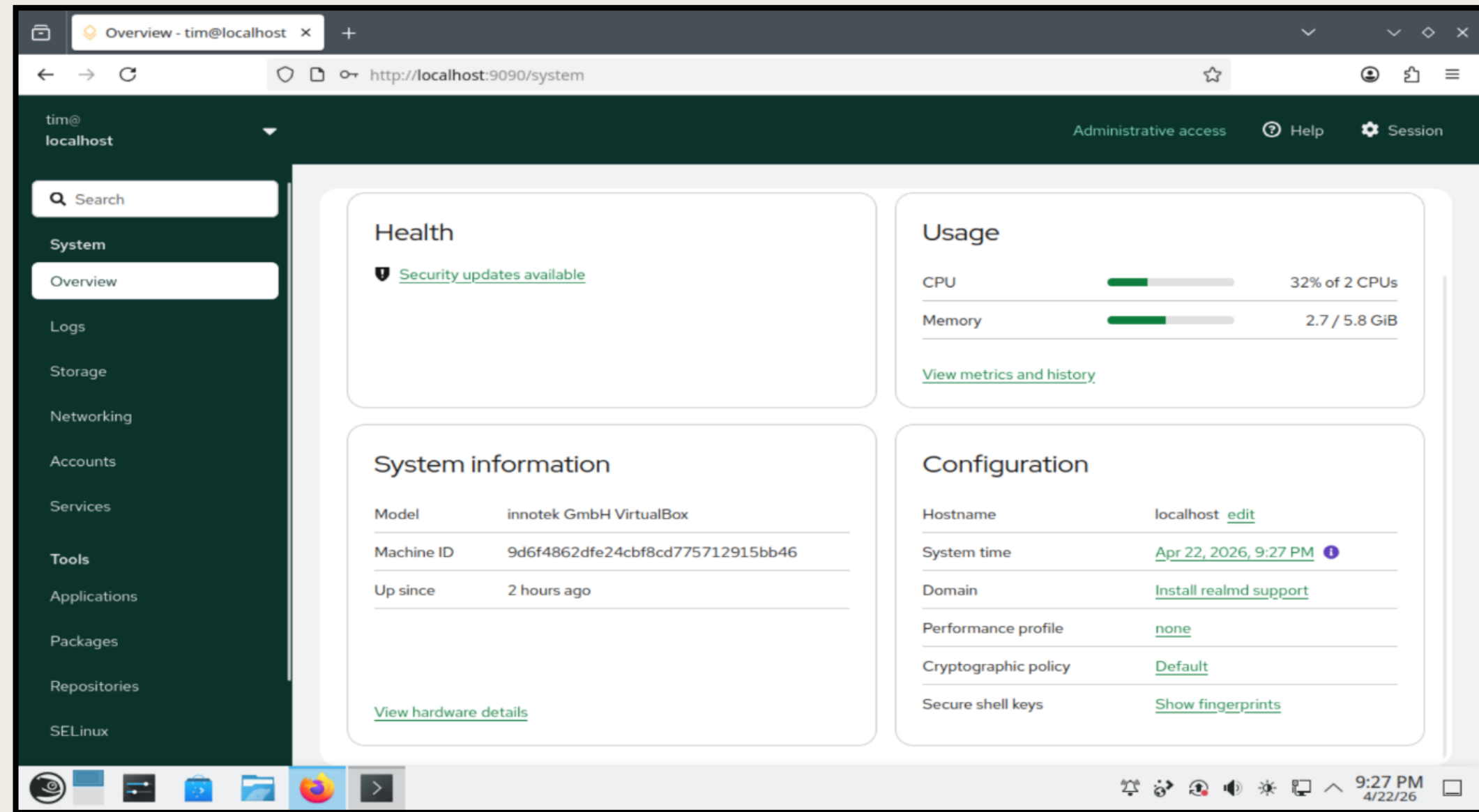
Myrlyn (App Installer)

- Open Myrlyn from the app launcher
- Search for any application by name
- Click **Install** → done.
- Clean, fast, focused



Key Functionality 3 – Memory Management

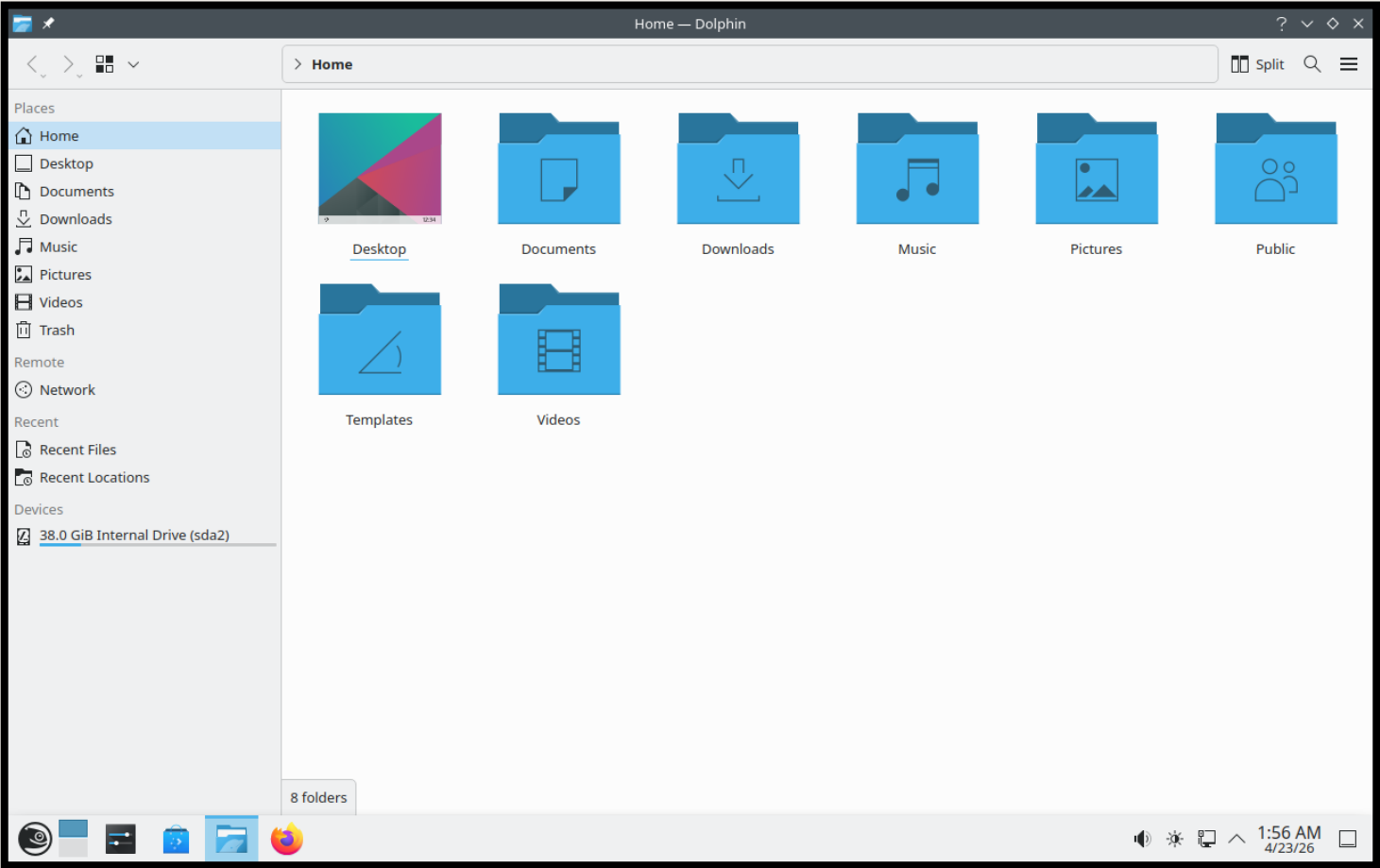
- Handled entirely by the **Linux kernel**
- **Virtual memory** — each running program gets its own protected memory space
- **Swap partition** acts as overflow when RAM is full
- **Cockpit** shows **real-time RAM usage** watch memory being allocated live as you open apps



Key Functionality 4 – Files, Storage Management

File System — Btrfs ("Better FS")

- Default filesystem in Leap 16
- Superpower: **automatic snapshots**
— saved before every update
- Btrfs = **Ctrl+Z** for your entire OS
- **Dolphin** is the default file manager
— tabbed, powerful, clean



```
~ : bash — Konsole
New Tab Split View
Copy Paste Find...
tim@localhost:~> sudo snapper list
[sudo] password for tim:

# | Type | Pre # | Date | User | Used Space | Cleanup | Description | Userdata
0 | single | | Wed 22 Apr 2026 06:59:44 PM IST | root | 188.02 MiB | | current |
1* | single | | Wed 22 Apr 2026 09:24:19 PM IST | root | 34.71 MiB | number | first root filesystem | important=no
2 | pre | | Wed 22 Apr 2026 09:25:11 PM IST | root | 15.16 MiB | number | zypp(zypper) | important=no
3 | post | 2 | | | | | |
tim@localhost:~>
```


Key Functionality 5 — Devices, I/O & Security

Device Management

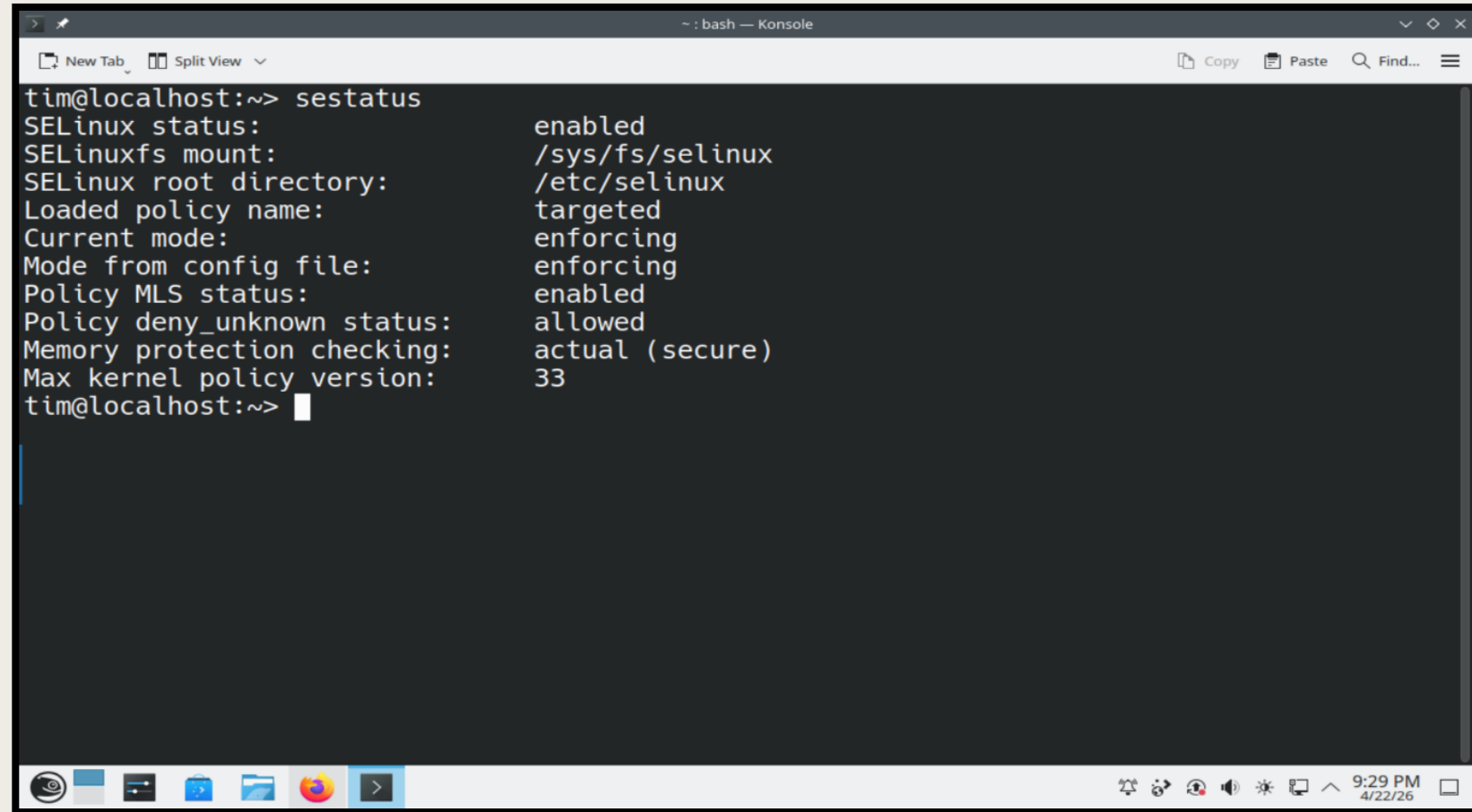
- Handled by **udev** — the Linux device manager
- Plug in a USB drive: udev detects it instantly, mounts it automatically
- Printers managed via **CUPS** (Common Unix Printing System)

I/O System

- All I/O flows through the **Linux kernel's I/O subsystem**
- Kernel treats devices, files, and network connections as **file descriptors** — one unified interface

Security — SELinux

- **SELinux** (Security-Enhanced Linux) enforced by default
- Originally developed by the **US National Security Agency (NSA)**
- Even if a program is hacked, SELinux **contains the damage**

A terminal window titled '~ : bash — Konsole' showing the output of the 'sestatus' command. The output lists various SELinux configuration details. The terminal has a dark background with light-colored text. The window includes a top bar with 'New Tab' and 'Split View' options, and a bottom bar with system icons and a clock showing 9:29 PM on 4/22/26.

```
tim@localhost:~> sestatus
SELinux status:                enabled
SELinuxfs mount:              /sys/fs/selinux
SELinux root directory:      /etc/selinux
Loaded policy name:           targeted
Current mode:                 enforcing
Mode from config file:       enforcing
Policy MLS status:           enabled
Policy deny_unknown status:  allowed
Memory protection checking:  actual (secure)
Max kernel policy version:    33
tim@localhost:~>
```

What openSUSE Leap Does Well — and What It Doesn't

Strengths

- **Rock-solid stability**
 - Rarely breaks — thoroughly tested before every release
- **Enterprise-grade security**
 - SELinux enforced by default — same as US DoD systems
- **Btrfs snapshots**
 - Roll back your entire OS like pressing Ctrl+Z
- **Long support lifecycle**
 - Leap 16 supported until 2031 — 6 years!
- **Cockpit web dashboard**
 - Manage your entire OS through a browser

Weaknesses

- **Steeper learning curve**
 - Not as beginner-friendly as Ubuntu or Mint
- **Smaller community**
 - Fewer online tutorials compared to Ubuntu
- **Not great for gaming**
 - SELinux causes friction; 32-bit disabled by default
- **Slower software updates**
 - Intentional, but newer app versions arrive late
- **Large download size**
 - Heavier ISO compared to minimal distros

History: Who Built It & When

